YEAR 7 · UNIT 1 · HOMEWORK 1

Integers and Cells

Mathematics 1.1 — Adding & Subtracting Integers
Science 1.1 — Cells and Microscopes

Name: ______________________ **Class:** ______________________

Date given: ____________________ **Date due:** ____________________

How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the calculation before the answer**.
- In science, answer in **full English sentences**. "Chloroplast" is not a sentence; "The green circles are chloroplasts" is.
- If a word is new to you, look for the blue **Word help** box on that page before you give up.

Section A · Mathematics 1.1 — Adding and Subtracting Integers

Remember from the lesson

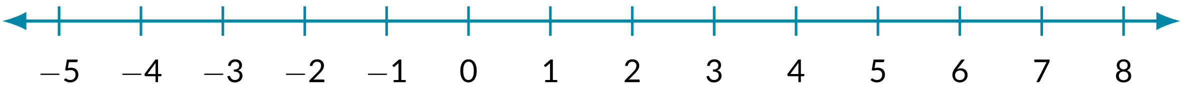
An **integer** is a whole number that is positive, negative or zero — never a fraction. On the number line, **negative** integers sit to the **left** of zero and **positive** integers to the **right**.

Add a **positive**	move **right**
Add a **negative**	move **left**
Subtract a **positive**	move **left**
Subtract a **negative**	move **right** — because $a - (-b) = a + b$

Careful: only $-(-)$ becomes $+$. A single $+(-)$ still sends you left, so $5 + (-3) = 2$, **not** 8.

A1 Show it on the number line

(a) Show $4 + (-3)$ on the number line below. Put a cross on the number you **start** at, then draw an arrow for the move.



$4 + (-3) =$ __________

(b) Complete each sentence with **higher than**, **lower than**, **warmer than** or **colder than**.

(i) -8 is ______________ -4.

(ii) A temperature of 3 °C is ______________ a temperature of -9 °C.

(iii) Floor -2 of the car park is ______________ floor 1.

A2 Moving along the line

Work these out. You may draw a small number line beside any of them.

(a) $-6 + 4 =$ ________

(b) $5 + (-9) =$ ________

(c) $-3 - 5 =$ ________

(d) $-2 - (-7) =$ ________

(e) $8 + (-8) =$ ________

(f) $-10 - (-4) =$ ________

(g) $0 - (-6) =$ ________

(h) $-7 + 15 =$ ________

(i) $4 - 11 =$ ________

(j) $-5 - (-5) =$ ________

(k) $6 - 8 - (-5) =$ ________

(l) $-9 - 6 + (-1) =$ ________

A3 Say it, then write it

Word help — these words decide the order of your calculation

subtract 5 from 8	you **start** at 8: $8 - 5$. The English gives the numbers in the **opposite order** to the calculation.
take 6 away from 2	the same trap: $2 - 6$.
7 less than 3	again the same trap: $3 - 7$.
rises by / falls by	the temperature goes up / the temperature goes down.
deposit	to put money **into** an account (+).
withdraw	to take money **out of** an account (−).
owe / a debt	money you must pay back. It makes your total **negative**.
descend	to go **down**. To go up is to **ascend**.

Turn each English sentence into a calculation, then work out the answer. **Write the calculation first.**

1. Subtract 9 from 4.
   Calculation: ______________________________ Answer: ___________

2. Take 6 away from -2.
   Calculation: ______________________________ Answer: ___________

3. What is 7 less than 3?
   Calculation: ______________________________ Answer: ___________

4. The temperature is $-8\,^{\circ}\mathrm{C}$ and it rises by 5 degrees.
   Calculation: ______________________________ Answer: ___________

5. A submarine is at -30 m. It descends a further 12 m.
   Calculation: ______________________________ Answer: ___________

6. An account holds -35 dollars. The bank removes a debt of 20 dollars from it.
   Calculation: ______________________________ Answer: ___________

A4 Difference

Remember: a difference is a gap

The **difference** between two numbers is how far apart they are on the line. Work it out with **bigger** $-$ **smaller**. A difference is **never negative**. **How much warmer**, **how much colder**, **how much higher** and **how many more** all ask for the same thing.

1. Find the difference between -6 and 2. Answer: ___________

2. Find the difference between -9 and -3. Answer: ___________

3. How much warmer is $4\,^{\circ}\mathrm{C}$ than $-13\,^{\circ}\mathrm{C}$? Show your calculation.

   ____

4. These two questions look almost the same. They are not.

   (a) Find the difference between -5 and 6. Answer: ___________

   (b) What is -5 minus 6? Answer: ___________

   (c) In one full sentence, explain why the two answers are different.

   ____

   ____

5. One winter morning it is $18\,^{\circ}\mathrm{C}$ in Ho Chi Minh City and $-24\,^{\circ}\mathrm{C}$ in Ulaanbaatar. How much colder is Ulaanbaatar? Show your calculation.

A5 Word problems

Read each one **all the way to the end** before you start writing. Show the calculation, then the answer.

1. **The office lift.** Mr Bowen gets into the lift on floor 7. He goes **down** 11 floors, and then **up** 2 floors. Which floor is he on now?

2. **A day in Sa Pa.** At 6 a.m. the temperature is $-3\,^\circ\mathrm{C}$. By noon it has risen 11 degrees. By midnight it has fallen 14 degrees.

   (a) What is the temperature at midnight?

   (b) What is the **difference** between the warmest and the coldest temperature of the day?

3. **Six weeks of saving.** Mr Bowen's account is at -40 dollars. Every Monday he deposits 25 dollars. Every Friday he withdraws 25 dollars. How much is in the account after 6 weeks? Explain your answer in one full sentence.

A6 Now you write the question

Your turn to be the teacher

So far the questions have given you the English, and you have written the maths. Now do it **backwards**. You are given the maths — write the English.

Your word problem must:

- be about something **real** — money, temperature, floors in a building, or depth under the sea;

- use at least one word from the **Word help** box on the earlier page, such as **rises by**, **falls by**, **deposits**, **withdraws**, **owes**, **below** or **descends**;
- be written in **full English sentences**, and end with a question;
- have the answer that is already given to you.

(a) Write a word problem for $-5 + 8 = 3$

(b) Write a word problem for $12 - (-4) = 16$

Section B · Science 1.1 — Cells and Microscopes

Remember from the lesson

A **cell** is the smallest basic unit of all living organisms. An **organelle** is a tiny structure inside a cell that does one specific, important job to keep the cell alive. Every cell has a **cell membrane**, **cytoplasm**, a **nucleus** and **mitochondria**. A plant cell has all of those **plus** a **cell wall** made of **cellulose**, **chloroplasts** containing **chlorophyll**, and a **sap vacuole**.

B1 Match the word to its job

Write the correct letter in the box next to each word.

Word	Letter	Job
1. Cell		**A** The control centre of the cell — the “boss”.
2. Organelle		**B** A large space of sugars and water that keeps a plant cell firm.
3. Nucleus		**C** Where energy is released from food.
4. Mitochondria		**D** The smallest basic unit of all living organisms.
5. Cell wall		**E** A tool that uses lenses to bend light and magnify an image.
6. Chloroplast		**F** A tiny structure inside a cell that does one specific job.
7. Sap vacuole		**G** A strong, stiff outer layer that holds a plant cell in shape.
8. Microscope		**H** Where a plant makes its food, using sunlight.

B2 Animal, plant, or both?

Put a tick (✓) in **one** column for each organelle.

Organelle	Animal cells only	Plant cells only	Both
Cell membrane			
Cytoplasm			
Nucleus			
Mitochondria			
Cell wall			
Chloroplast			
Sap vacuole			

B3 Answer in full sentences

Sentence starters — you may copy these

“A plant cell needs a cell wall because…”
“One difference is that… Another difference is that…”
“A microscope is a tool that…”

1. An animal has a skeleton to hold it up. A plant does not. Explain why every plant cell needs a cell wall.

2. Give **two** ways a cell wall is different from a cell membrane. Write each one as a full sentence.

3. Under the microscope, a leaf cell is full of small green circles. Name them, and name the green substance inside them.

4. What does a microscope do? Use the words **lens** and **magnify** in your answer.

B4 A model of a cell

Mr Bowen builds a model of an animal cell in his kitchen. He uses a clear plastic bag, filled with jelly, with a plum and four red beans pushed into the middle of it.

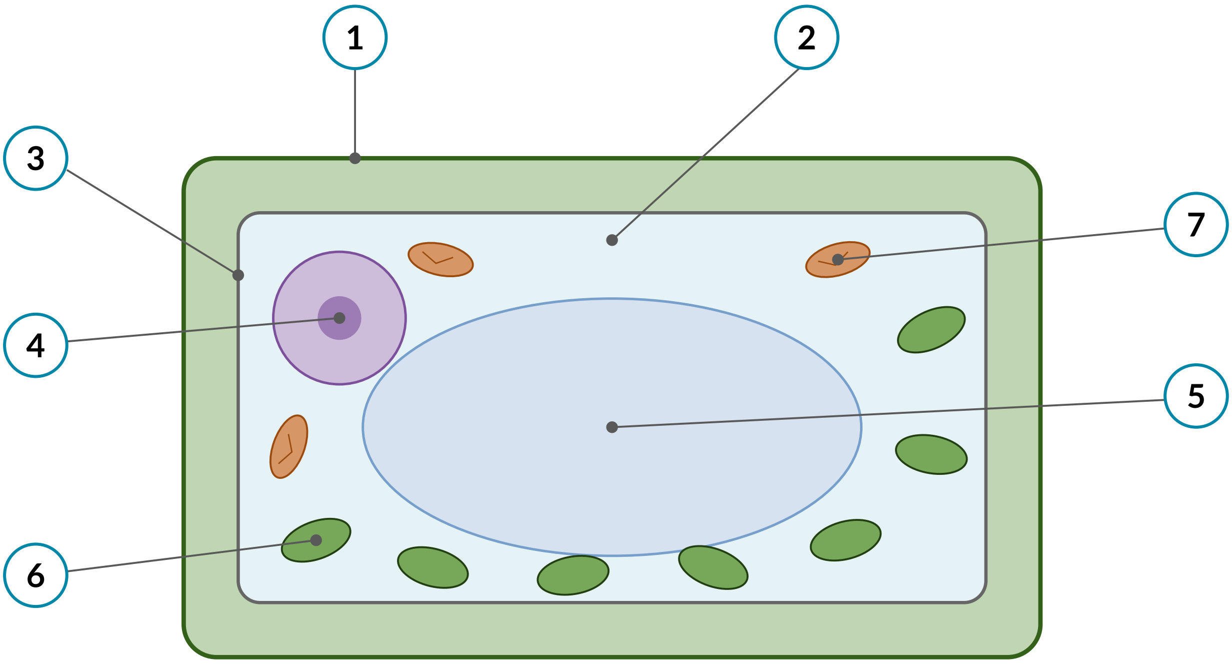
(a) Which organelle does each item represent?

Item in the model	Organelle it represents
The plastic bag	
The jelly	
The plum	
The four beans	

(b) Name **one** thing he could add to turn it into a model of a **plant** cell, and say which organelle it would represent.

B5 Label the plant cell

Write the name of each numbered part on the lines below the diagram.



1. ______________________________

2. ______________________________

3. ______________________________

4. ______________________________

5. ______________________________

6. ______________________________

7. ______________________________

Hint: number 3 is the very thin layer just inside the outer green edge. Numbers 1 and 3 are easy to confuse — one is thick and stiff, one is thin and flexible.